

Supplemental Material: Atomic Coherence of 2 minutes and Instability of 1.5×10^{-18} at 1 s in a Wannier-Stark Lattice Clock

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I. DDTWA CALCULATION FOR WANNIER-STARK LATTICE CLOCK

In the main text, we used numerical simulations based on the dissipative discrete truncated Wigner approximation (DDTWA) for modeling the experimental data. Our method combines DTWA [1] with quantum jumps, which is similar to Ref. [2] but is generalized beyond two-level atoms. An alternative method using stochastic differential equations is discussed in Ref. [3].

Firstly, we briefly explain the quantum jump method. We consider the Lindblad master equation,

$$\frac{d}{dt}\hat{\rho} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] + \sum_{j=1}^{n_{\text{tot}}} \left(\hat{L}_j \hat{\rho} \hat{L}_j^\dagger - \frac{1}{2}(\hat{L}_j^\dagger \hat{L}_j \hat{\rho} + \hat{\rho} \hat{L}_j^\dagger \hat{L}_j) \right), \quad (1)$$

where $\hat{H}$ is the Hamiltonian, and $\hat{L}_j$ are the quantum jump operators. We define the non-Hermitian Hamiltonian

$$\hat{H}_{\text{NH}} = \hat{H} - \frac{i\hbar}{2} \sum_{j=1}^{n_{\text{tot}}} \hat{L}_j^\dagger \hat{L}_j. \quad (2)$$

The quantum jump method for solving Lindblad master equation can be described in the following way:

- Choose a random number r_1 based on the uniform distribution between 0 and 1. Starting from a pure state $|\psi(0)\rangle$, evolve the wave function under the non-Hermitian Hamiltonian $\hat{H}_{\text{NH}}$,

$$|\psi^{(1)}(\tau)\rangle = \exp\left(-i\frac{\hat{H}_{\text{NH}}\tau}{\hbar}\right)|\psi(0)\rangle. \quad (3)$$

A quantum jump occurs at time τ such that $\langle\psi^{(1)}(\tau)|\psi^{(1)}(\tau)\rangle = r_1$.

- Choose a random number r_2 based on the uniform distribution between 0 and 1. Then, we choose the jump operator $\hat{L}_n$ in such a way that n is the smallest integer satisfying

$$\sum_{j=1}^n P_j(\tau) \geq r_2, \quad P_j(\tau) = \frac{\langle\psi^{(1)}(\tau)|\hat{L}_j^\dagger \hat{L}_j|\psi^{(1)}(\tau)\rangle}{\sum_{j=1}^{n_{\text{tot}}} \langle\psi^{(1)}(\tau)|\hat{L}_j^\dagger \hat{L}_j|\psi^{(1)}(\tau)\rangle}, \quad (4)$$

and the wave function becomes

$$|\psi(\tau)\rangle = \frac{\hat{L}_n|\psi^{(1)}(\tau)\rangle}{\sqrt{\langle\psi^{(1)}(\tau)|\hat{L}_n^\dagger \hat{L}_n|\psi^{(1)}(\tau)\rangle}}. \quad (5)$$

- Repeat the procedure above with $|\psi(\tau)\rangle$ as the new initial state until the final simulation time is reached.

To combine with DTWA, we modify the procedure above in the following way:

- Replace the wave function evolution under the non-Hermitian Hamiltonian $\hat{H}_{\text{NH}}$ with DTWA evolution for a set of single-atom operators $\hat{O}_k$ and the identity operator $\hat{I}$. This leads to mean-field equations for expectation values $\langle\psi^{(1)}|\hat{O}_k|\psi^{(1)}\rangle$ and the wave function norm $\langle\psi^{(1)}|\psi^{(1)}\rangle$, with random sampling of initial conditions [1, 4]. The random sampling of initial conditions is based on the following rules: Suppose $\hat{O}_k$ has eigenvalues $\alpha_j^{(k)}$ with corresponding eigenvectors $|\alpha_j^{(k)}\rangle$, we set $\langle\psi^{(1)}(0)|\hat{O}_k|\psi^{(1)}(0)\rangle = \alpha_j^{(k)}$ with probability $P_j^{(k)} = |\langle\psi^{(1)}(0)|\alpha_j^{(k)}\rangle|^2$. Note that we always have $\langle\psi^{(1)}(0)|\psi^{(1)}(0)\rangle = 1$.
- When a quantum jump occurs, we reset the initial conditions based on the selected quantum jump operator and then perform DTWA evolution. We repeat the procedure until the final simulation time is reached.

Now we apply the method described above to numerically simulate the dynamics in the Wannier-Stark lattice clock. Following the discussions in Ref. [5], we have

$$\hat{H} = \hat{H}_{\text{twolevel}} + \hat{H}_{\text{spectator}}, \quad (6)$$

where

$$\hat{H}_{\text{twolevel}} = \sum_{\substack{\mathbf{nm} \\ (\mathbf{n} \neq \mathbf{m})}} \left[J_{\mathbf{nm}} \hat{\mathbf{S}}_{\mathbf{n}} \cdot \hat{\mathbf{S}}_{\mathbf{m}} + \chi_{\mathbf{nm}} \hat{S}_{\mathbf{n}}^z \hat{S}_{\mathbf{m}}^z + D_{\mathbf{nm}} (\hat{S}_{\mathbf{n}}^x \hat{S}_{\mathbf{m}}^y - \hat{S}_{\mathbf{n}}^y \hat{S}_{\mathbf{m}}^x) + \frac{C_{\mathbf{nm}}}{2} (\hat{S}_{\mathbf{n}}^z \hat{N}_{\mathbf{m}}^{eg} + \hat{N}_{\mathbf{n}}^{eg} \hat{S}_{\mathbf{m}}^z) \right], \quad (7)$$

$$\hat{H}_{\text{spectator}} = \sum_{\substack{\mathbf{nm} \\ (\mathbf{n} \neq \mathbf{m})}} \left[\frac{K_{\mathbf{nm}}^a}{2} (\hat{S}_{\mathbf{n}}^z \hat{N}_{\mathbf{m}}^a + \hat{N}_{\mathbf{n}}^a \hat{S}_{\mathbf{m}}^z) + \frac{K_{\mathbf{nm}}^b}{2} (\hat{S}_{\mathbf{n}}^z \hat{N}_{\mathbf{m}}^b + \hat{N}_{\mathbf{n}}^b \hat{S}_{\mathbf{m}}^z) \right]. \quad (8)$$

Here, $\mathbf{n} = \{n_x, n_y, n\}$, with n_x, n_y are the number of radial harmonic oscillator modes and n is the index of lattice sites where the Wannier-Stark state centered at. $\hat{H}_{\text{twolevel}}$ couples between the clock states $|e\rangle \equiv |^3P_0, m_F = -3/2\rangle$ and $|g\rangle \equiv |^1S_0, m_F = -5/2\rangle$. These are the two levels that define the spin-1/2 operators $\hat{S}_{\mathbf{n}}^{x,y,z}$. They are defined in an energy lattice where $\mathbf{n}$ denotes an occupied mode $\mathbf{n}$, and $\hat{N}_{\mathbf{n}}^{eg}$ describes the sum of population in $|e\rangle$ and $|g\rangle$ states for mode $\mathbf{n}$. We only include occupied modes in the summation, and they are sampled from thermal distribution based on the cloud temperature. For simplicity, we drop all the terms in the Hamiltonian acting identically on $|g\rangle$ and $|e\rangle$ states since they have no contributions to the clock coherence and stability.

The Hamiltonian parameters in $\hat{H}_{\text{twolevel}}$ have already been discussed in Ref. [5], where we include contributions from both on-site p -wave interactions and nearest-neighbor s -wave interactions. For completeness, here we list them as follows:

$$\begin{aligned} J_{\mathbf{nm}} &= (V_{\mathbf{nm}}^{eg} - U_{\mathbf{nm}}^{eg}) \cos[(n - m)\varphi], \\ \chi_{\mathbf{nm}} &= (V_{\mathbf{nm}}^{ee} + V_{\mathbf{nm}}^{gg} - 2V_{\mathbf{nm}}^{eg}) + (V_{\mathbf{nm}}^{eg} - U_{\mathbf{nm}}^{eg}) (1 - \cos[(n - m)\varphi]), \\ D_{\mathbf{nm}} &= -(V_{\mathbf{nm}}^{eg} - U_{\mathbf{nm}}^{eg}) \sin[(n - m)\varphi], \\ C_{\mathbf{nm}} &= V_{\mathbf{nm}}^{ee} - V_{\mathbf{nm}}^{gg}, \end{aligned} \quad (9)$$

where $\varphi = \pi\lambda_L/\lambda_c$ is the clock laser phase difference between nearest-neighbor sites (λ_L is the lattice wavelength, λ_c is the clock laser wavelength), $U_{\mathbf{nm}}^{\alpha\beta}$ and $V_{\mathbf{nm}}^{\alpha\beta}$ are s -wave and p -wave interaction parameters respectively ($\alpha, \beta = \{g, e\}$),

$$U_{\mathbf{nm}}^{\alpha\beta} = \frac{4\pi\hbar a_{\alpha\beta}}{M} s_{n_x m_x} s_{n_y m_y} \eta_{nm}, \quad V_{\mathbf{nm}}^{\alpha\beta} = \frac{3\pi\hbar b_{\alpha\beta}^3}{M} (p_{n_x m_x} s_{n_y m_y} + s_{n_x m_x} p_{n_y m_y}) \eta_{nm}. \quad (10)$$

Here, M is the mass of ^{87}Sr atoms, $a_{\alpha\beta}$ and $b_{\alpha\beta}^3$ are elastic s -wave scattering lengths and elastic p -wave scattering volumes with $a_{eg} \equiv (a_{eg}^+ + a_{eg}^-)/2$ and $b_{eg}^3 \equiv [(b_{eg}^+)^3 + (b_{eg}^-)^3]/2$ (the $+$ and $-$ sign in the superscript of scattering lengths and scattering volumes labels the triplet and singlet scattering channel respectively, see Ref. [5] for details). s_{nm} , p_{nm} and η_{nm} are the wave-function overlap integrals, $s_{nm} = \int dX [\phi_n(X)]^2 [\phi_m(X)]^2$, $p_{nm} = \int dX [(\partial_X \phi_n(X)) \phi_m(X) - \phi_n(X) (\partial_X \phi_m(X))]^2$, $\eta_{nm} = \int dZ [W_n(Z)]^2 [W_m(Z)]^2$, where $\phi_n(X)$ are harmonic oscillator wave functions, and $W_n(Z)$ are Wannier-Stark wave functions.

$\hat{H}_{\text{spectator}}$ describes the interactions between atoms in clock states ($|e\rangle$ or $|g\rangle$) and spectator atoms (not in clock states). Light scattering from the lattice beams and spontaneous emission both can induce decay from $|e\rangle$ to $|g\rangle$ during clock interrogation. After a decay event, atoms can populate other nuclear spins in the ground state manifold. This leads to a non-negligible on-site s -wave density shift. Here we use $|a\rangle$ to label a ground state level (different from $|g\rangle$) that has the same nuclear spin as the excited state $|e\rangle$. The interactions between $|a\rangle$ and $|e\rangle$ ($|g\rangle$) states can be described by s -wave scattering length a_{eg}^- (a_{gg}). We use $|b\rangle$ to label all other ground state levels (different from $|g\rangle$ and $|a\rangle$) with different nuclear spins compared to the excited state $|e\rangle$. Notice that $|b\rangle$ might refer to several ground state levels, and there is no need to distinguish them because the interaction would be the same. The interactions between $|b\rangle$ and $|e\rangle$ ($|g\rangle$) states can be described by s -wave scattering length a_{eg} (a_{gg}). We use operators $\hat{N}_{\mathbf{n}}^a$ and $\hat{N}_{\mathbf{n}}^b$ to describe the number of atoms in mode $\mathbf{n}$ with state $|a\rangle$ and $|b\rangle$, respectively. The Hamiltonian parameters relevant for $\hat{H}_{\text{spectator}}$ are given by

$$K_{\mathbf{nm}}^a = U_{\mathbf{nm}}^{eg-} - U_{\mathbf{nm}}^{gg}, \quad K_{\mathbf{nm}}^b = U_{\mathbf{nm}}^{eg} - U_{\mathbf{nm}}^{gg}. \quad (11)$$

Apart from the Hamiltonian dynamics, we include all the dissipative terms discussed in Fig. 1 of the main text. We define fermionic annihilation operator $\hat{c}_{\mathbf{n}\alpha}$ for mode $\mathbf{n}$ and internal state α . The Lindblad jump operators are listed below:

- Spontaneous emission and lattice-induced decay

$$\sqrt{\Gamma_{eg}} \hat{c}_{\mathbf{n}g}^\dagger \hat{c}_{\mathbf{n}e}, \quad \sqrt{\Gamma_{ea}} \hat{c}_{\mathbf{n}a}^\dagger \hat{c}_{\mathbf{n}e}, \quad \sqrt{\Gamma_{eb}} \hat{c}_{\mathbf{n}b}^\dagger \hat{c}_{\mathbf{n}e}, \quad \forall \mathbf{n} \quad (12)$$

Here we assume the decay processes do not change the motional state of the atoms. This approximation, therefore, neglects recoil heating. Γ_{eg} , Γ_{ea} , Γ_{eb} can be calculated from $\Gamma_L(0)$ and $\partial_U \Gamma_L$ in the Table I of the main text by multiplying the branching ratio for different nuclear spin states.

- Single-body loss

$$\sqrt{\Gamma_e} \hat{c}_{\mathbf{n}e}, \quad \sqrt{\Gamma_g} \hat{c}_{\mathbf{n}g}, \quad \sqrt{\Gamma_g} \hat{c}_{\mathbf{n}a}, \quad \sqrt{\Gamma_g} \hat{c}_{\mathbf{n}b}, \quad \forall \mathbf{n} \quad (13)$$

Γ_e and Γ_g are taken from the Table I of the main text.

- Two-body ee loss

$$\sqrt{\Gamma_{\mathbf{nm}}} \hat{c}_{\mathbf{n}e} \hat{c}_{\mathbf{m}e}, \quad \forall \mathbf{n}, \mathbf{m}, \quad (14)$$

where (β_{ee}^3 is the inelastic p -wave scattering volume)

$$\Gamma_{\mathbf{nm}} = \frac{3\pi\hbar^2\beta_{ee}^3}{2M}(p_{n_x m_x} s_{n_y m_y} + s_{n_x m_x} p_{n_y m_y})\eta_{nm}. \quad (15)$$

The thermal average of $\Gamma_{\mathbf{nm}}$ agrees with K_{ee}/v in Table I of the main text, where v is the effective volume set by the density.

In the description above, we simply treated the lattice-induced scattering to 3P_2 states as atom loss. In the experiment, atoms can also stay in the long-lived 3P_2 states and increase the strength of spectator interaction. However, the scattering lengths of the 3P_2 states are unknown. To account for this effect, we consider a rescaling of spectator interaction $\hat{H}_{\text{spectator}} \rightarrow \xi \hat{H}_{\text{spectator}}$, with ξ a fitting parameter. We choose $\xi = 3$ for since this value generates the best description of the experiment.

In Fig. S1, we compare the DDTWA simulations and the experimental data. Both show a clear dependence on the atom number per site (N_{site}) and exhibit a slight nonlinearity. To account for this nonlinearity in our linear extrapolation in the main text, we perform fits over two different ranges of N_{site} and include the difference between these fits as an additional source of uncertainty. Experimentally observed spatial inhomogeneity in the contrast decay rate is presumably due to small spatial inhomogeneity in the initial spin distribution. This could bias the analysis and complicates the modeling. Instead, we use half of the region with better spin distribution for the analysis in the main text.

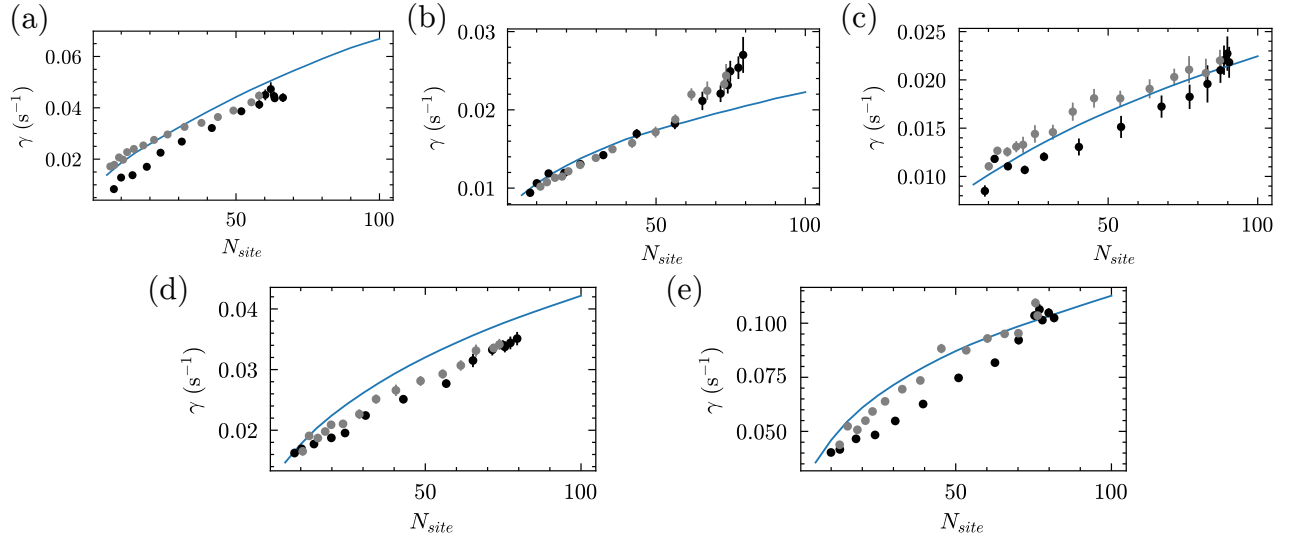


FIG. S1. N_{site} dependence of the contrast decay rate γ for at (a) $5E_r$, (b) $8E_r$, (c) $11E_r$, (d) $23E_r$, and (e) $69E_r$. The black markers represent the regions with higher initial spin purity, and the gray markers show the other part of the sample. The error bars show 68% confidence interval. The blue solid lines are DDTWA simulation.

II. LOWER BOUND OF SENSITIVITY FOR ELLIPSE FITTING

In ellipse fitting, we measure the excitation fraction x_i, y_i in two separate regions ($i = 1, \dots, n$ is labeling independent measurements), and use this information to determine the

differential phase ϕ . The expectation value of x_i, y_i is given by

$$E(x_i) = \frac{C}{2} \cos(\theta_i - \phi/2) + D, \quad E(y_i) = \frac{C}{2} \cos(\theta_i + \phi/2) + D. \quad (16)$$

In particular, we focus on the case where the common phase θ_i is randomly distributed between 0 and 2π due to laser phase noise, so the values (x_i, y_i) form an ellipse. Here, C is the Ramsey contrast, and D is the center of the ellipse. The conditional probability for the repeated measurements is given by

$$p(\mathbf{x}, \mathbf{y} | \phi, \boldsymbol{\theta}) = \prod_i p(x_i, y_i | \phi, \theta_i), \quad (17)$$

where $p(x_i, y_i | \phi, \theta_i)$ is the probability of measurement outcome x_i, y_i conditioned on parameters ϕ and θ_i . In principle, x_i, y_i are also conditioned on parameters C and D , here we omit them to shorten the notations. For simplicity, we assume the quantum state is separable between the two regions, and also the state in each region can be approximated by a Gaussian state, leading to

$$p(x_i, y_i | \phi, \theta_i) = p(x_i | \phi, \theta_i) p(y_i | \phi, \theta_i) \\ \approx \frac{1}{\sqrt{2\pi \text{Var}(x_i)}} \exp\left(-\frac{1}{2} \frac{(x_i - E(x_i))^2}{\text{Var}(x_i)}\right) \times \frac{1}{\sqrt{2\pi \text{Var}(y_i)}} \exp\left(-\frac{1}{2} \frac{(y_i - E(y_i))^2}{\text{Var}(y_i)}\right), \quad (18)$$

in which $\text{Var}(x_i)$ depends on $\theta_i - \phi/2$, and $\text{Var}(y_i)$ depends on $\theta_i + \phi/2$.

The lower bound for the variance of the differential phase ϕ can be estimated by the classical Cramér–Rao bound. A straightforward way to apply the classical Cramér–Rao bound is to integrate out the common phases θ_i [6],

$$p(\mathbf{x}, \mathbf{y} | \phi) = \prod_i p(x_i, y_i | \phi), \quad p(x_i, y_i | \phi) = \int_0^{2\pi} \frac{d\theta_i}{2\pi} p(x_i, y_i | \phi, \theta_i), \quad (19)$$

and then construct the classical Fisher information,

$$F_\phi = \int d\mathbf{x} d\mathbf{y} p(\mathbf{x}, \mathbf{y} | \phi) \left(\frac{\partial \ln p(\mathbf{x}, \mathbf{y} | \phi)}{\partial \phi} \right)^2 = n \int dx dy p(x, y | \phi) \left(\frac{\partial \ln p(x, y | \phi)}{\partial \phi} \right)^2, \quad (20)$$

in which the last step is based on our assumption of independent measurements. The classical Cramér–Rao bound leads to

$$\text{Var}(\phi) \geq F_\phi^{-1}. \quad (21)$$

This approach provides the tightest lower bound for $\text{Var}(\phi)$, while analytical calculation for this bound is hard to achieve.

Here we would like to provide a slightly looser lower bound for $\text{Var}(\phi)$, where we can have more analytical insight. We start from the classical Fisher information matrix for all the parameters $\phi, \boldsymbol{\theta}$,

$$F = \begin{pmatrix} F_{\phi\phi} & F_{\phi\theta_1} & \cdots & F_{\phi\theta_n} \\ F_{\theta_1\phi} & F_{\theta_1\theta_1} & \cdots & F_{\theta_1\theta_n} \\ \vdots & \vdots & \ddots & \vdots \\ F_{\theta_n\phi} & F_{\theta_n\theta_1} & \cdots & F_{\theta_n\theta_n} \end{pmatrix}, \quad (22)$$

where

$$F_{\phi\phi} = \int d\mathbf{x}d\mathbf{y} p(\mathbf{x}, \mathbf{y}|\phi, \boldsymbol{\theta}) \left(\frac{\partial \ln p(\mathbf{x}, \mathbf{y}|\phi, \boldsymbol{\theta})}{\partial \phi} \right)^2 \approx \sum_i \left(\frac{(\partial_\phi E(x_i))^2}{\text{Var}(x_i)} + \frac{(\partial_\phi E(y_i))^2}{\text{Var}(y_i)} \right). \quad (23)$$

$$F_{\phi\theta_i} = F_{\theta_i\phi} = \int d\mathbf{x}d\mathbf{y} p(\mathbf{x}, \mathbf{y}|\phi, \boldsymbol{\theta}) \left(\frac{\partial \ln p(\mathbf{x}, \mathbf{y}|\phi, \boldsymbol{\theta})}{\partial \phi} \right) \left(\frac{\partial \ln p(\mathbf{x}, \mathbf{y}|\phi, \boldsymbol{\theta})}{\partial \theta_i} \right) \approx \frac{(\partial_\phi E(x_i))(\partial_{\theta_i} E(x_i))}{\text{Var}(x_i)} + \frac{(\partial_\phi E(y_i))(\partial_{\theta_i} E(y_i))}{\text{Var}(y_i)}, \quad (24)$$

$$F_{\theta_i\theta_j} = \int d\mathbf{x}d\mathbf{y} p(\mathbf{x}, \mathbf{y}|\phi, \boldsymbol{\theta}) \left(\frac{\partial \ln p(\mathbf{x}, \mathbf{y}|\phi, \boldsymbol{\theta})}{\partial \theta_i} \right) \left(\frac{\partial \ln p(\mathbf{x}, \mathbf{y}|\phi, \boldsymbol{\theta})}{\partial \theta_j} \right) \approx \left(\frac{(\partial_{\theta_i} E(x_i))^2}{\text{Var}(x_i)} + \frac{(\partial_{\theta_i} E(y_i))^2}{\text{Var}(y_i)} \right) \delta_{ij}. \quad (25)$$

The calculation above is based on the nature of the Gaussian state and the large atom number limit in each region, so we can drop the derivatives for $\text{Var}(x_i)$ and $\text{Var}(y_i)$.

Based on the multivariate classical Cramér–Rao bound and the fact that the diagonal element of a positive semi-definite matrix is positive semi-definite, we have

$$\text{Var}(\phi) \geq (F^{-1})_{\phi\phi} = \left(F_{\phi\phi} - \sum_i F_{\phi\theta_i} F_{\theta_i\theta_i}^{-1} F_{\theta_i\phi} \right)^{-1}. \quad (26)$$

Here the last step is based on the inverse of a block matrix. Notice that $\partial_{\theta_i} E(x_i) = -2\partial_\phi E(x_i)$, $\partial_{\theta_i} E(y_i) = 2\partial_\phi E(y_i)$, we have

$$\begin{aligned} F_{\phi\phi} - \sum_i F_{\phi\theta_i} F_{\theta_i\theta_i}^{-1} F_{\theta_i\phi} &\approx 4 \sum_i \left(\frac{(\partial_\phi E(x_i))^2}{\text{Var}(x_i)} + \frac{(\partial_\phi E(y_i))^2}{\text{Var}(y_i)} \right)^{-1} \frac{(\partial_\phi E(x_i))^2}{\text{Var}(x_i)} \frac{(\partial_\phi E(y_i))^2}{\text{Var}(y_i)} \\ &= \frac{C^2}{4} \sum_i \frac{1}{\csc^2(\theta_i - \phi/2) \text{Var}(x_i) + \csc^2(\theta_i + \phi/2) \text{Var}(y_i)} \\ &= \frac{nC^2}{4} \int_0^{2\pi} \frac{d\theta}{2\pi} \frac{1}{\csc^2(\theta - \phi/2) \text{Var}(x) + \csc^2(\theta + \phi/2) \text{Var}(y)}, \end{aligned} \quad (27)$$

in which $\text{Var}(x)$ and $\text{Var}(y)$ are functions of θ and ϕ . So we have

$$\text{Var}(\phi) \geq \frac{4}{nC^2} \left[\int_0^{2\pi} \frac{d\theta}{2\pi} \left(\frac{1}{\csc^2(\theta - \phi/2) \text{Var}(x) + \csc^2(\theta + \phi/2) \text{Var}(y)} \right) \right]^{-1}. \quad (28)$$

Note that this formula agrees with the formula listed in the Supplemental Material of Ref. [7].

Now we would like to benchmark the two different lower bounds (see Eq. (21) and Eq. (28)). We consider the quantum state is a product state with contrast $C = 0.9$, so the variances for excitation fraction x and y are given by $\text{Var}(x) = E(x)(1 - E(x))/N_a$, $\text{Var}(y) = E(y)(1 - E(y))/N_a$, where N_a is the atom number for each region. In Fig. S2, we compare Eq. (21) and Eq. (28) for $N_a = 20$ and $N_a = 200$. We find that Eq. (21) and Eq. (28) agree for most of the ϕ except a small region near $\phi = 0$, and the region of disagreement is shrinking as we increase N_a . Considering $N_a \sim 10^5$ in our experiment, we believe that Eq. (28) is a good approximation for Eq. (21) except for very small ϕ .

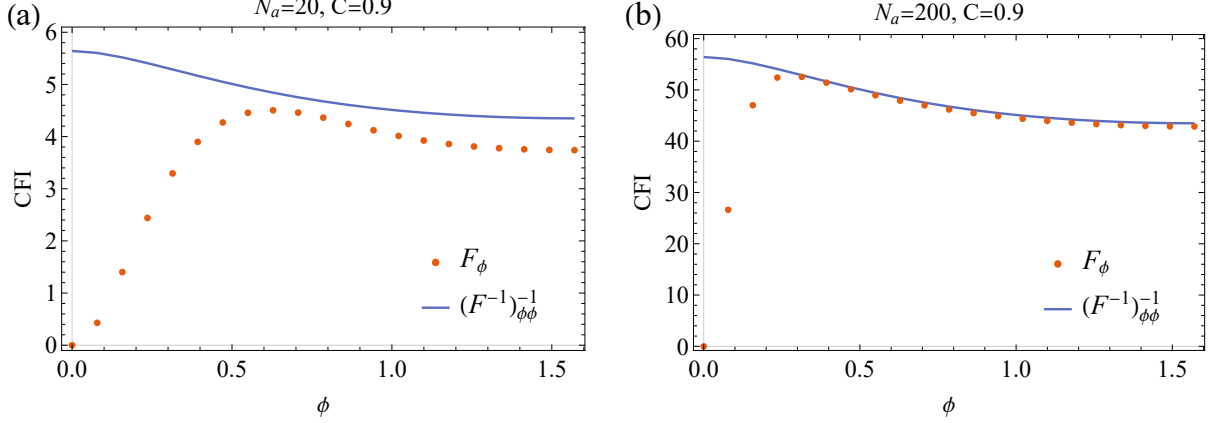


FIG. S2. Comparison Eq. (21) and Eq. (28) for differential phase ϕ with (a) $N_a = 20$ and (b) $N_a = 200$. We consider the quantum state is a product state with contrast $C = 0.9$.

A physical interpretation for $F_\phi \sim 0$ near $\phi = 0$ can be gained by noticing that the ellipse is becoming a diagonal line, and the quantum noise makes the data points to be either above or below that line. In that case $\phi > 0$ and $\phi < 0$ cases cannot be distinguished well and therefore we are not extracting any information using this method. Since the quantum noise will decrease as we increase N_a , the value of ϕ for which we can distinguish whether the data points are above or below the diagonal line decreases with increasing N_a .

III. ELLIPSE FITTING METHOD

Extracting the differential phase ϕ from the ellipse fitting can be done in various ways [6, 8–10]. We employ two different methods: (1) least squares with geometric distance (LSGD) and (2) Bayesian inference (BI). For estimating uncertainty, we use the jackknife method [7] for both LSGD and BI. BI with Markov chain Monte Carlo (MCMC) provides an approximate posterior distribution of ϕ [11]. We use the standard deviation of the marginalized posterior distribution as the uncertainty and compare it with jackknife method.

LSGD uses a cost function based on the sum of geometric distances

$$\text{Cost} = \sum_{i=1}^n d_i = \sum_{i=1}^n (x_i - E(x_i))^2 + (y_i - E(y_i))^2, \quad (29)$$

where d_i is the geometric distance between the data point (x_i, y_i) with index $i = 1, 2, \dots, n$ and the points on an ellipse, and fitted values $(E(x_i), E(y_i))$ are from Eq. (16) for a given $n + 4$ parameters $\{C, \phi, \mathbf{D}, \boldsymbol{\theta}\}$ ($\mathbf{D}$ is separated for x, y).

For BI, we model the data points $\{(x_i, y_i)\}$ as

$$(x_i, y_i) | C, \phi, \mathbf{D}, \theta_i, N \sim (\mathcal{N}(E(x_i), \sigma_{x_i}^2(N_a)), \mathcal{N}(E(y_i), \sigma_{y_i}^2(N_a))), \quad (30)$$

where $\mathcal{N}(\mu, \sigma^2)$ is the normal distribution with mean μ and variance σ^2 . $\sigma_{x_i}^2(N_a) = x_i(1 - x_i)/N_a$ is a variance estimation assuming large-number limit of a coherent spin state, and we also infer an effective atom number N_a per ensemble from the data. This approximation is only valid for a sufficiently large atom number [12]. For the prior distribution, we first perform a LSGD fit to the data, and use the fitted parameters as the mean values of the

normally distributed prior. Prior sensitivity is checked by varying the prior width for a given data set. The posterior distribution is insensitive to width changes from 0.05 to 0.0001 for C , ϕ , and θ for the data set used in the main text and the simulated data presented here.

Estimating the uncertainty of ϕ using the jackknife method is done by removing one data point at index i and performing the LS GD or BI to the remaining data points [7]. The bias corrected estimator for ϕ for each i is given by

$$\phi_i^{\text{JK}} = n\bar{\phi}^{\text{JK}} - (n-1)\bar{\phi}_{\neq i}^{\text{JK}}, \quad (31)$$

where $\bar{\phi}^{\text{JK}}$ is the ϕ obtained from using all data points, and $\bar{\phi}_{\neq i}^{\text{JK}}$ is obtained from using all data points except i .

We check the performance of the fitting methods with simulated data assuming a coherent spin state and compare with the result from the experimental data.

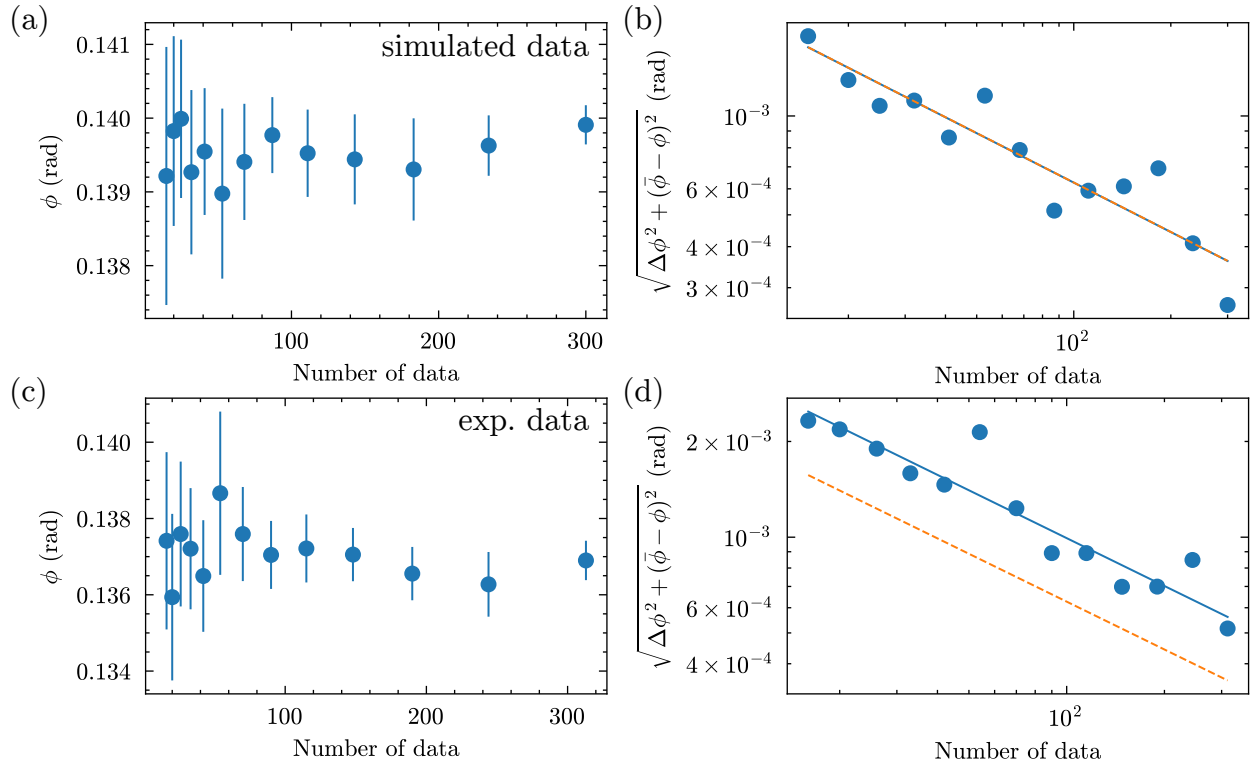


FIG. S3. Bayesian inference of ϕ with increasing data set size from product state simulation (a, b) and from experimental data (c, d). (a) Estimated ϕ as a function of the number of data points. The marker represents the mean value of the posterior distribution, and the error bar represents the 68% confidence interval. (b) Uncertainty of the ϕ as a function of the number of data points. The blue solid line is a fit ($\propto 1/\sqrt{n}$) to the data and the orange dashed line is the theoretical estimation (6.3 mrad for 1 data point). $\bar{\phi}$ is ϕ with the full data set, used for estimating the bias. (c, d) are the same as (a, b) but with experimental data. The experimental data show 1.6 times larger noise (the blue solid line).

Figure S3 shows the results of the BI for ϕ with increasing data set size. We simulate data assuming $\{C = 0.74, \phi = 0.14, N = 160,000, D = 0.5\}$, parameters close to our experimental conditions, with θ_i randomly sampled from a uniform distribution between 0 and 2π . For the theoretical estimation, we use Eq. (28). For the simulated data, we find that the stability

obtained from the posterior distribution of ϕ roughly matches the theoretical prediction. When the same method is applied to experimental data, it shows larger uncertainty. The scatter in the variance likely arises from the finite sample size.

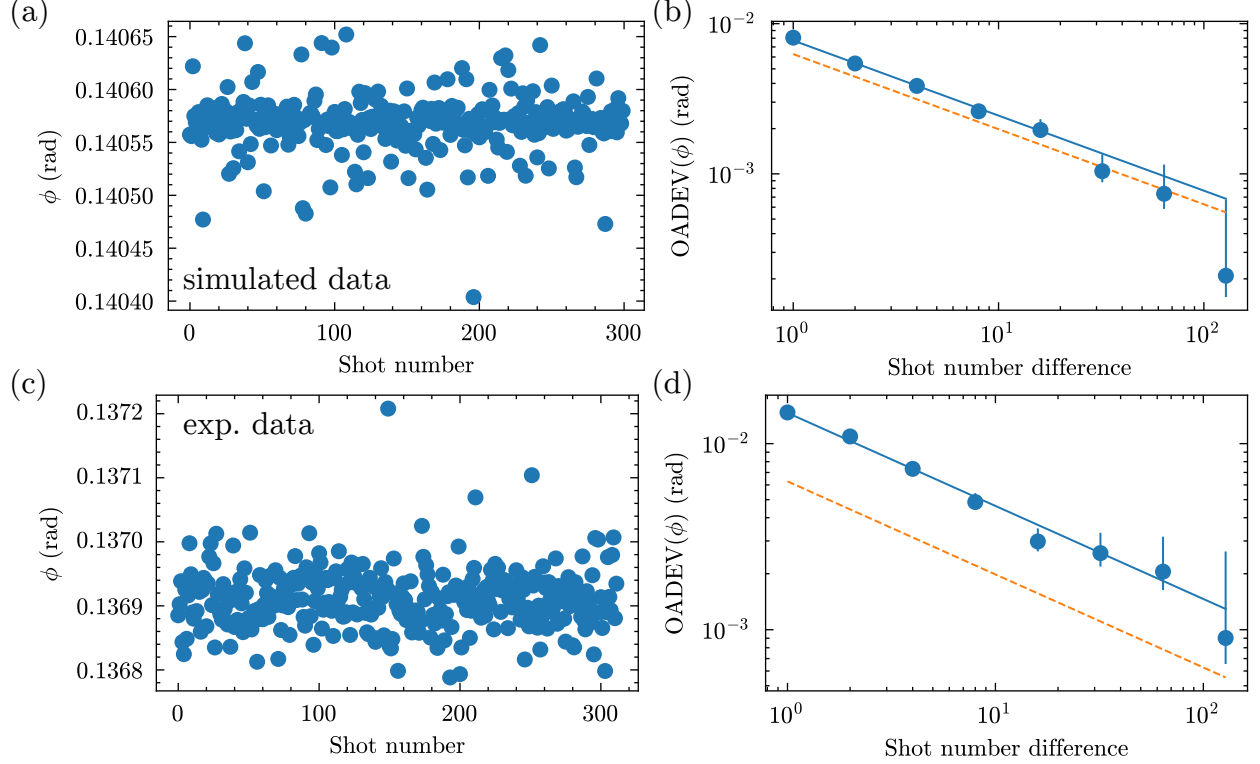


FIG. S4. Bayesian inference with the jackknife method using data set from product state simulation (a, b) and from experimental data (c, d). (a) Estimated ϕ from Eq. (31). (b) Allan deviation of ϕ . The blue solid line is a fit to the data and the orange dashed line is the theoretical estimation (6.3 mrad for 1 data). (c, d) are the same as (a, b) but with experimental data. Experimental data shows 2.4 times larger noise (the blue solid line).

Figure S4 shows the jackknife estimation with BI. For the simulated data set, the jackknife method gives results similar to the theoretical estimation.

Figure S5 shows the jackknife estimation with LSGD, which yields results similar to the BI method. The corresponding results for experimental data are shown in the main text as Fig. 3(e).

We have generally found good agreement between different methods, with the remaining discrepancies likely arising from the finite sample size of $n \sim 300$. The DDTWA simulation indicates a higher noise level compared to simulations based on coherent spin states, which is primarily due to the stochastic generation of spectator atoms (statistical mixture). It is thus challenging to develop a noise model for Bayesian analysis using DDTWA simulated data. Consequently, we decided to use the most straightforward method, LSGD with jackknife, for the analysis presented in the main text.

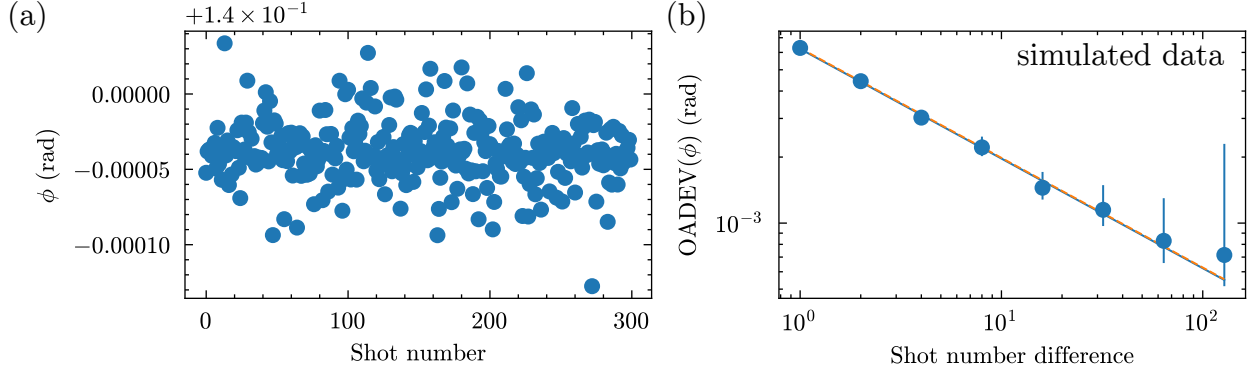


FIG. S5. Jackknife estimation with LSGD for a product state simulation. (a) Estimated ϕ from Eq. (31). (b) Allan deviation of ϕ . The blue solid line is a fit to the data and the orange dashed line is the theoretical estimation (6.3 mrad for 1 data point). The results for experimental data are shown in the main text.

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- [1] J. Schachenmayer, A. Pikovski, and A. M. Rey, Many-body quantum spin dynamics with monte carlo trajectories on a discrete phase space, *Phys. Rev. X* **5**, 011022 (2015).
 - [2] V. P. Singh and H. Weimer, Driven-dissipative criticality within the discrete truncated wigner approximation, *Phys. Rev. Lett.* **128**, 200602 (2022).
 - [3] J. Huber, A. M. Rey, and P. Rabl, Realistic simulations of spin squeezing and cooperative coupling effects in large ensembles of interacting two-level systems, *Phys. Rev. A* **105**, 013716 (2022).
 - [4] B. Zhu, A. M. Rey, and J. Schachenmayer, A generalized phase space approach for solving quantum spin dynamics, *New Journal of Physics* **21**, 082001 (2019).
 - [5] A. Aepli, A. Chu, T. Bothwell, C. J. Kennedy, D. Kedar, P. He, A. M. Rey, and J. Ye, Hamiltonian engineering of spin-orbit-coupled fermions in a Wannier-Stark optical lattice clock, *Science Advances* **8**, eadc9242 (2022).
 - [6] R. Corgier, M. Malitesta, L. A. Sidorenkov, F. P. D. Santos, G. Rosi, G. M. Tino, A. Smerzi, L. Salvi, and L. Pezzè, Squeezing-enhanced accurate differential sensing under large phase noise (2025), arXiv:2501.18256 [quant-ph].
 - [7] G. E. Marti, R. B. Hutson, A. Goban, S. L. Campbell, N. Poli, and J. Ye, Imaging Optical Frequencies with 100 μ Hz Precision and 1.1 μ m Resolution, *Physical Review Letters* **120**, 103201 (2018).
 - [8] J. K. Stockton, X. Wu, and M. A. Kasevich, Bayesian estimation of differential interferometer phase, *Physical Review A* **76**, 033613 (2007).
 - [9] G. T. Foster, J. B. Fixler, J. M. McGuirk, and M. A. Kasevich, Method of phase extraction between coupled atom interferometers using ellipse-specific fitting, *Optics Letters* **27**, 951 (2002).
 - [10] M. J. Collett and L. R. Watkins, Ellipse fitting for interferometry Part 3: Dynamic method, *Journal of the Optical Society of America A* **32**, 491 (2015).
 - [11] O. Abril-Pla, V. Andreani, C. Carroll, L. Dong, C. J. Fongesbeck, M. Kochurov, R. Kumar, J. Lao, C. C. Luhmann, O. A. Martin, M. Osthege, R. Vieira, T. Wiecki, and R. Zinkov, PyMC: A modern, and comprehensive probabilistic programming framework in Python, *PeerJ Computer Science* **9**, e1516 (2023).
 - [12] W. J. Eckner, N. Darkwah Oppong, A. Cao, A. W. Young, W. R. Milner, J. M. Robinson, J. Ye, and A. M. Kaufman, Realizing spin squeezing with Rydberg interactions in an optical clock, *Nature* **621**, 734 (2023).